Sources: Sheldon Axler, *Measure, Integration & Real Analysis* (MIRA) and its *Supplement*, at measure.axler.net.

Exercise 4.1 (MIRA 5B.1).

(a) Let λ denote Lebesgue measure on $[0,1]$. Show that

$$\int_{[0,1]} \int_{[0,1]} \frac{x^2 - y^2}{(x^2+y^2)^2}\, d\lambda(y)\, d\lambda(x) = \frac{\pi}{4}$$

and

$$\int_{[0,1]} \int_{[0,1]} \frac{x^2 - y^2}{(x^2+y^2)^2}\, d\lambda(x)\, d\lambda(y) = -\frac{\pi}{4}.$$

(b) Explain why (a) violates neither Tonelli's Theorem nor Fubini's Theorem.

Solution 4.1. Here λ is Lebesgue measure on $([0,1], \mathcal{B})$, where $\mathcal{B}$ is the σ-algebra of Borel subsets of $[0,1]$. The quotient in the statement is not defined at $(0,0)$; define $f\colon [0,1] \times [0,1] \to \mathbf{R}$ by $f(x,y) = (x^2-y^2)/(x^2+y^2)^2$ for $(x,y) \neq (0,0)$ and $f(0,0) = 0$. Let $g(x) = 1/(1+x^2)$ for $x \in [0,1]$.

(a) Suppose $x \in (0,1]$. Because $x^2+y^2 \geq x^2 > 0$ for every $y \in [0,1]$, the function $y \mapsto f(x,y)$ is continuous on $[0,1]$, and it is the derivative of $y \mapsto y/(x^2+y^2)$. It is bounded, because a continuous function on a closed bounded set is bounded, and it is Riemann integrable because continuous functions are Riemann integrable (1.11). Thus the equality of the Riemann and Lebesgue integrals (3.34) and the fundamental theorem of calculus give

$$\int_{[0,1]} f(x,y)\, d\lambda(y) = \int_0^1 f(x,y)\, dy = \frac{1}{1+x^2} = g(x).$$

In the same way, if $y \in (0,1]$, then $x \mapsto f(x,y)$ is continuous on $[0,1]$ and is the derivative of $x \mapsto -x/(x^2+y^2)$, and thus $\int_{[0,1]} f(x,y)\, d\lambda(x) = -1/(1+y^2) = -g(y)$. Also, g is continuous on $[0,1]$ and is the derivative of $\arctan$, and thus 3.34 gives $\int_{[0,1]} g\, d\lambda = \int_0^1 g(x)\, dx = \arctan 1 - \arctan 0 = \pi/4$.

The function $y \mapsto f(0,y)$ is continuous on $(0,1]$, and thus it is Borel measurable, because every continuous function is Borel measurable (2.41) and $\{0\}$ is a Borel set. Because $f(0,y) \leq 0$ for every $y \in [0,1]$, its integral is defined and equals $-c$ for some $c \in [0,\infty]$, by the definition of the integral of a real-valued function (3.18). Thus $G(x) = \int_{[0,1]} f(x,y)\, d\lambda(y)$ defines a function $G\colon [0,1] \to [-\infty,\infty]$ that equals the continuous function g on $(0,1]$ and $-c$ at 0. Hence G is $\mathcal{B}$-measurable for the same reason as $y \mapsto f(0,y)$, with $G^+ = \chi_{(0,1]} g$ and $G^- = c\chi_{\{0\}}$. The integral of a simple function (3.7) gives $\int \chi_{\{0\}}\, d\lambda = \lambda(\{0\}) = 0$ and $\int G^-\, d\lambda = c\lambda(\{0\}) = 0$, where $\infty \cdot 0 = 0$ if $c = \infty$. Because $g = G^+ + \chi_{\{0\}}$ on $[0,1]$, additivity of integration for nonnegative functions (3.16) gives $\int G^+\, d\lambda = \int_{[0,1]} g\, d\lambda = \pi/4$. Hence 3.18 gives

$$\int_{[0,1]} \int_{[0,1]} f(x,y)\, d\lambda(y)\, d\lambda(x) = \int G^+\, d\lambda - \int G^-\, d\lambda = \frac{\pi}{4}.$$

Similarly, $H(y) = \int_{[0,1]} f(x,y)\, d\lambda(x)$ defines a $\mathcal{B}$-measurable function $H\colon [0,1] \to [-\infty,\infty]$ that equals $-g$ on $(0,1]$, and $H(0) \in [0,\infty]$ because $f(x,0) \geq 0$ for every $x \in [0,1]$. Hence $H^+ = H(0)\chi_{\{0\}}$ and $H^- = \chi_{(0,1]} g = G^+$. Thus $\int H^+\, d\lambda = 0$ by 3.7, and 3.18 gives

$$\int_{[0,1]} \int_{[0,1]} f(x,y)\, d\lambda(x)\, d\lambda(y) = \int H^+\, d\lambda - \int H^-\, d\lambda = -\frac{\pi}{4}.$$

(b) The functions $(x,y) \mapsto x$ and $(x,y) \mapsto y$ on $[0,1] \times [0,1]$ are $\mathcal{B} \otimes \mathcal{B}$-measurable, because the inverse images of a Borel set $A \subseteq \mathbf{R}$ under them are the measurable rectangles $(A \cap [0,1]) \times [0,1]$ and $[0,1] \times (A \cap [0,1])$. Let $E = \{0\} \times \{0\}$, which is also a measurable rectangle. Then

$$f(x,y) = \frac{x^2 - y^2}{(x^2+y^2)^2 + \chi_E(x,y)}$$

for all $(x,y) \in [0,1] \times [0,1]$, and the denominator is never 0. Thus f is $\mathcal{B} \otimes \mathcal{B}$-measurable by algebraic operations with measurable functions (2.46), and hence so is $|f|$.

Tonelli's Theorem (5.28) requires a function with values in $[0,\infty]$, and $f(x,y) < 0$ whenever $0 \le x < y \le 1$; thus it does not apply to f. It does apply to $|f|$, because λ is finite and hence σ-finite, and gives $\int |f| \, d(\lambda \times \lambda) = \int_{[0,1]} I \, d\lambda$, where $I \colon [0,1] \to [0,\infty]$ is the $\mathcal{B}$-measurable function defined by $I(x) = \int_{[0,1]} |f(x,y)| \, d\lambda(y)$. Suppose $x \in (0,1]$. Because $f(x,y) \ge 0$ for $y \in [0,x]$, we have $|f(x,y)| \ge \chi_{[0,x]}(y) f(x,y) \ge 0$ for every $y \in [0,1]$. Thus order preservation of integration (3.8) and the definition of integration on a subset (3.24) give $I(x) \ge \int_{[0,x]} f(x,y) \, d\lambda(y)$. The function $y \mapsto f(x,y)$ is continuous on $[0,x]$ with the antiderivative of (a), and thus 3.34 gives

$$I(x) \ge \int_0^x f(x,y) \, dy = \frac{x}{x^2 + x^2} = \frac{1}{2x}.$$

Now suppose $n \in \mathbf{Z}^+$. If $k \in \{1, \ldots, n\}$ and $x \in [4^{-k}, 2 \cdot 4^{-k}]$, then $I(x) \ge 1/(2x) \ge 4^{k-1}$. The intervals $[4^{-k}, 2 \cdot 4^{-k}]$ for $k = 1, \ldots, n$ are subsets of $[0,1]$ that are disjoint because $2 \cdot 4^{-k-1} < 4^{-k}$, and the kth has Lebesgue measure 4^{-k} because the outer measure of a closed interval is its length (2.14). Thus $I \ge \sum_{k=1}^n 4^{k-1} \chi_{[4^{-k}, 2 \cdot 4^{-k}]}$ on $[0,1]$, and 3.8 and 3.7 give

$$\int |f| \, d(\lambda \times \lambda) = \int_{[0,1]} I \, d\lambda \ge \sum_{k=1}^n 4^{k-1} \cdot 4^{-k} = \frac{n}{4}.$$

Because this holds for every $n \in \mathbf{Z}^+$, we have $\int |f| \, d(\lambda \times \lambda) = \infty$. Thus f does not satisfy the hypothesis $\int |f| \, d(\lambda \times \lambda) < \infty$ of Fubini's Theorem (5.32). Because neither theorem applies to f, the unequal iterated integrals in (a) violate neither. □

Exercise 4.2 (MIRA 5B.2).

(a) Give an example of a doubly indexed collection $\{x_{m,n} : m, n \in \mathbf{Z}^+\}$ of real numbers such that

$$\sum_{m=1}^\infty \sum_{n=1}^\infty x_{m,n} = 0 \quad \text{and} \quad \sum_{n=1}^\infty \sum_{m=1}^\infty x_{m,n} = \infty.$$

(b) Explain why (a) violates neither Tonelli's Theorem nor Fubini's Theorem.

Solution 4.2. (a) For $m, n \in \mathbf{Z}^+$, let

$$x_{m,n} = \begin{cases} m & \text{if } n = m, \\ -m & \text{if } n = m+1, \\ 0 & \text{otherwise.} \end{cases}$$

Suppose $m \in \mathbf{Z}^+$. The only nonzero terms of $\sum_{n=1}^\infty x_{m,n}$ are $x_{m,m} = m$ and $x_{m,m+1} = -m$, and thus $\sum_{n=1}^\infty x_{m,n} = m - m = 0$. Hence $\sum_{m=1}^\infty \sum_{n=1}^\infty x_{m,n} = \sum_{m=1}^\infty 0 = 0$. Now suppose $n \in \mathbf{Z}^+$. If $n = 1$, then the only nonzero term of $\sum_{m=1}^\infty x_{m,n}$ is $x_{1,1} = 1$. If $n \ge 2$, then the

only nonzero terms are $x_{n-1,n} = -(n-1)$ and $x_{n,n} = n$. In both cases $\sum_{m=1}^{\infty} x_{m,n} = 1$. Thus $\sum_{n=1}^{\infty}\sum_{m=1}^{\infty} x_{m,n} = \sum_{n=1}^{\infty} 1 = \infty$, because $\sum_{n=1}^{N} 1 = N$ for every $N \in \mathbf{Z}^+$.

(b) Let $\mathcal{S}$ be the σ-algebra of all subsets of $\mathbf{Z}^+$, let μ be counting measure on $(\mathbf{Z}^+, \mathcal{S})$, and define $f\colon \mathbf{Z}^+ \times \mathbf{Z}^+ \to \mathbf{R}$ by $f(m,n) = x_{m,n}$. Integration of a nonnegative sequence with respect to μ is summation (Example 3.6). By the definition of the integral of a real-valued function (3.18), the same holds for each sequence $n \mapsto x_{m,n}$ and each sequence $m \mapsto x_{m,n}$, because its positive and negative parts each have at most one nonzero term. Hence the two equations in (a) state that

$$\int_{\mathbf{Z}^+}\int_{\mathbf{Z}^+} f(m,n)\,d\mu(n)\,d\mu(m) = 0 \quad \text{and} \quad \int_{\mathbf{Z}^+}\int_{\mathbf{Z}^+} f(m,n)\,d\mu(m)\,d\mu(n) = \infty.$$

Tonelli's Theorem (5.28) and Fubini's Theorem (5.32) each conclude that these two iterated integrals are equal for a function that satisfies the hypotheses of the theorem.

The measure μ is σ-finite, because $\mathbf{Z}^+ = \bigcup_{k=1}^{\infty}\{k\}$ and $\mu(\{k\}) = 1$ for every $k \in \mathbf{Z}^+$. Every subset E of $\mathbf{Z}^+ \times \mathbf{Z}^+$ is in $\mathcal{S}\otimes\mathcal{S}$, because E is the union over $m \in \mathbf{Z}^+$ of the measurable rectangles $\{m\} \times \{n \in \mathbf{Z}^+ : (m,n) \in E\}$. Thus f and $|f|$ are $\mathcal{S}\otimes\mathcal{S}$-measurable. However, 5.28 requires a function with values in $[0,\infty]$, and $f(1,2) = -1$. Next, 5.32 requires $\int_{\mathbf{Z}^+\times\mathbf{Z}^+} |f|\,d(\mu\times\mu) < \infty$. Applying 5.28 to the nonnegative function $|f|$ and then 3.6 to the inner and outer integrals, we have

$$\int_{\mathbf{Z}^+\times\mathbf{Z}^+} |f|\,d(\mu\times\mu) = \sum_{m=1}^{\infty}\sum_{n=1}^{\infty}|x_{m,n}| = \sum_{m=1}^{\infty} 2m \geq \sum_{m=1}^{\infty} 1 = \infty.$$

Thus f satisfies the hypotheses of neither theorem, and (a) violates neither. □

Exercise 4.3 (MIRA 6C.1). Show that the map $f \mapsto \|f\|$ from a normed vector space V to $\mathbf{F}$ is continuous (where the norm on $\mathbf{F}$ is the usual absolute value).

Solution 4.3. Suppose $f, g \in V$. The triangle inequality in the definition of a norm (6.33) gives $\|f\| = \|(f-g)+g\| \leq \|f-g\| + \|g\|$, and thus $\|f\| - \|g\| \leq \|f-g\|$. Interchanging f and g and using the homogeneity of the norm, we have $\|g\| - \|f\| \leq \|g-f\| = \|(-1)(f-g)\| = |-1|\,\|f-g\| = \|f-g\|$. Because $\big|\|f\| - \|g\|\big|$ equals $\|f\| - \|g\|$ or $\|g\| - \|f\|$, we conclude that $\big|\|f\| - \|g\|\big| \leq \|f-g\|$ for all $f, g \in V$.

Because normed vector spaces are metric spaces (6.36), the distance between two elements of V or of $\mathbf{F}$ is the norm of their difference. Now suppose $f \in V$ and $\varepsilon > 0$. If $g \in V$ and $\|f-g\| < \varepsilon$, then the inequality above gives $\big|\|f\| - \|g\|\big| \leq \|f-g\| < \varepsilon$. Thus the definition of continuity (6.10) is satisfied at each $f \in V$ with $\delta = \varepsilon$, completing the proof. □

Exercise 4.4 (MIRA 6C.13). Suppose U is a normed vector space. Let d be the metric on U defined by $d(f,g) = \|f-g\|$ for $f, g \in U$. Let V be the complete metric space constructed in Exercise 16 in Section 6A.

(a) Show that the set V is a vector space under natural operations of addition and scalar multiplication.

(b) Show that there is a natural way to make V into a normed vector space and that with this norm, V is a Banach space.

(c) Explain why (b) shows that every normed vector space is a subspace of some Banach space.

Solution 4.4. We write (f_k) for a sequence $(f_1, f_2, \dots)$ in U. In the construction of V, the set W consists of the Cauchy sequences in U, and $(f_k) \equiv (g_k)$ means that $\lim_{k\to\infty} d(f_k, g_k) = 0$. The elements of V are the equivalence classes $(f_k)\widehat{\ }$ of the sequences $(f_k) \in W$, and $d_V\big((f_k)\widehat{\ }, (g_k)\widehat{\ }\big) = \lim_{k\to\infty} d(f_k, g_k)$.

(a) Suppose $(f_k), (g_k) \in W$ and $\alpha \in \mathbf{F}$. To see that $(f_k + g_k)$ and (αf_k) are Cauchy sequences, suppose $\varepsilon > 0$ and let $\delta = \varepsilon/(2 + |\alpha|)$; then $2\delta \le \varepsilon$ and $|\alpha|\delta < \varepsilon$. By the definition of a Cauchy sequence (6.12), applied to (f_k) and to (g_k), there exists $n \in \mathbf{Z}^+$ such that $\|f_j - f_k\| < \delta$ and $\|g_j - g_k\| < \delta$ for all integers $j \ge n$ and $k \ge n$. For such j and k, the definition of a norm (6.33) gives

$$\|(f_j + g_j) - (f_k + g_k)\| \le \|f_j - f_k\| + \|g_j - g_k\| < 2\delta \le \varepsilon,$$
$$\|\alpha f_j - \alpha f_k\| = |\alpha|\,\|f_j - f_k\| \le |\alpha|\delta < \varepsilon.$$

Thus $(f_k + g_k) \in W$ and $(\alpha f_k) \in W$. Now suppose $(u_k), (v_k) \in W$ with $(f_k) \equiv (u_k)$ and $(g_k) \equiv (v_k)$. Then $(u_k + v_k) \in W$ and $(\alpha u_k) \in W$ by what we just proved. Suppose $\varepsilon > 0$ and let δ be as above. Because $\lim_{k\to\infty}\|f_k - u_k\| = 0$ and $\lim_{k\to\infty}\|g_k - v_k\| = 0$, there exists $m \in \mathbf{Z}^+$ such that $\|f_k - u_k\| < \delta$ and $\|g_k - v_k\| < \delta$ for all integers $k \ge m$. For such k, we have $\|(f_k + g_k) - (u_k + v_k)\| \le \|f_k - u_k\| + \|g_k - v_k\| < \varepsilon$ and $\|\alpha f_k - \alpha u_k\| = |\alpha|\,\|f_k - u_k\| < \varepsilon$. Thus $(f_k + g_k) \equiv (u_k + v_k)$ and $(\alpha f_k) \equiv (\alpha u_k)$. Hence the equations

$$(f_k)\widehat{\ } + (g_k)\widehat{\ } = (f_k + g_k)\widehat{\ } \quad \text{and} \quad \alpha(f_k)\widehat{\ } = (\alpha f_k)\widehat{\ }$$

define an addition and a scalar multiplication on V, because their right sides do not depend on the sequences chosen from the equivalence classes on the left.

The operations of V are computed termwise on sequences chosen from the equivalence classes. Thus commutativity, associativity, the multiplicative identity property and the distributive properties in the definition of a vector space (6.27) hold in V because they hold in U for the kth terms of those sequences for every $k \in \mathbf{Z}^+$; for example, $\alpha\big((f_k)\widehat{\ } + (g_k)\widehat{\ }\big) = \big(\alpha(f_k + g_k)\big)\widehat{\ } = (\alpha f_k + \alpha g_k)\widehat{\ } = \alpha(f_k)\widehat{\ } + \alpha(g_k)\widehat{\ }$. The sequence $(0, 0, \dots)$ is a Cauchy sequence because $d(0, 0) = 0$, and $(f_k)\widehat{\ } + (0, 0, \dots)\widehat{\ } = (f_k + 0)\widehat{\ } = (f_k)\widehat{\ }$; thus $(0, 0, \dots)\widehat{\ }$ is an additive identity of V, which we denote by 0. Finally, $(-f_k) = \big((-1)f_k\big) \in W$ and $(f_k)\widehat{\ } + (-f_k)\widehat{\ } = \big(f_k + (-f_k)\big)\widehat{\ } = 0$; thus $(-f_k)\widehat{\ }$ is an additive inverse of $(f_k)\widehat{\ }$. Hence V is a vector space.

(b) Define $\|\cdot\|_V \colon V \to [0, \infty)$ by $\|\varphi\|_V = d_V(\varphi, 0)$. Suppose $\varphi, \psi \in V$ and $\alpha \in \mathbf{F}$, and let $(f_k), (g_k) \in W$ be such that $\varphi = (f_k)\widehat{\ }$ and $\psi = (g_k)\widehat{\ }$. Because $0 = (0, 0, \dots)\widehat{\ }$ and $d(f_k, 0) = \|f_k\|$ for every $k \in \mathbf{Z}^+$, the construction of V gives $\|\varphi\|_V = \lim_{k\to\infty}\|f_k\|$. Because d_V is a metric, the definition of a metric (6.1) shows that $\|\varphi\|_V = 0$ if and only if $\varphi = 0$. It also gives

$$\|\varphi + \psi\|_V = d_V(\varphi + \psi, 0) \le d_V(\varphi + \psi, \psi) + d_V(\psi, 0) = \|\varphi\|_V + \|\psi\|_V,$$

where $d_V(\varphi + \psi, \psi) = \lim_{k\to\infty}\|(f_k + g_k) - g_k\| = \lim_{k\to\infty}\|f_k\| = \|\varphi\|_V$. To prove homogeneity, let $L = \|\varphi\|_V$ and suppose $\varepsilon > 0$. Because $\lim_{k\to\infty}\|f_k\| = L$, there exists $m \in \mathbf{Z}^+$ such that $\big|\|f_k\| - L\big| < \varepsilon/(1 + |\alpha|)$ for all integers $k \ge m$. For such k, 6.33 gives $\big|\|\alpha f_k\| - |\alpha|L\big| = |\alpha|\,\big|\|f_k\| - L\big| < \varepsilon$. Because $\alpha\varphi = (\alpha f_k)\widehat{\ }$, this shows that $\|\alpha\varphi\|_V = \lim_{k\to\infty}\|\alpha f_k\| = |\alpha|\,\|\varphi\|_V$. Hence $\|\cdot\|_V$ satisfies the three conditions of 6.33, and $(V, \|\cdot\|_V)$ is a normed vector space.

Because $(-g_k)\widehat{\ }$ is the additive inverse of ψ by (a), we have $\varphi - \psi = (f_k - g_k)\widehat{\ }$, and thus

$$\|\varphi - \psi\|_V = \lim_{k\to\infty}\|f_k - g_k\| = \lim_{k\to\infty} d(f_k, g_k) = d_V(\varphi, \psi).$$

Because normed vector spaces are metric spaces (6.36) with the metric $(\varphi, \psi) \mapsto \|\varphi - \psi\|_V$, the equation above shows that the metric of the normed vector space V is d_V. The metric space

(V, d_V) is complete by the construction of V; hence V is a complete normed vector space, which is the definition of a Banach space (6.37).

(c) Define $T\colon U \to V$ by $Tf = (f, f, \dots)\widehat{\ }$, where $(f, f, \dots) \in W$ because $d(f, f) = 0$. Suppose $f, g \in U$ and $\alpha \in \mathbf{F}$. The definitions in (a) give $T(f+g) = (f+g, f+g, \dots)\widehat{\ } = Tf+Tg$ and $T(\alpha f) = (\alpha f, \alpha f, \dots)\widehat{\ } = \alpha Tf$; thus T is linear. By (b), we have $\|Tf\|_V = \lim_{k\to\infty}\|f\| = \|f\|$. If $Tf = Tg$, then $\|f - g\| = \|T(f - g)\|_V = \|Tf - Tg\|_V = 0$, and thus $f = g$. Hence T is one-to-one. The range $T(U)$ contains $T0 = 0$, and it is closed under addition and scalar multiplication because $Tf + Tg = T(f + g)$ and $\alpha Tf = T(\alpha f)$; thus $T(U)$ is a subspace of V by the conditions for a subspace (6.31).

Because T is one-to-one, linear and norm preserving, identifying each $f \in U$ with $Tf \in V$ changes neither the vector space operations nor the norm of U, and it makes U the subspace $T(U)$ of the Banach space V. Thus every normed vector space is, after this identification, a subspace of some Banach space. □

Exercise 4.5 (MIRA 7A.1). Suppose μ is a measure. Prove that

$$\|f + g\|_\infty \le \|f\|_\infty + \|g\|_\infty \quad \text{and} \quad \|\alpha f\|_\infty = |\alpha|\,\|f\|_\infty$$

for all $f, g \in \mathcal{L}^\infty(\mu)$ and all $\alpha \in \mathbf{F}$. Conclude that with the usual operations of addition and scalar multiplication of functions, $\mathcal{L}^\infty(\mu)$ is a vector space.

Solution 4.5. Let $(X, \mathcal{S}, \mu)$ denote the measure space on which μ is defined. For each $\mathcal{S}$-measurable function $h\colon X \to \mathbf{F}$, let $T_h = \{t > 0 : \mu(\{x \in X : |h(x)| > t\}) = 0\}$; thus $\|h\|_\infty = \inf T_h$ by the definition of the essential supremum (7.1). Each set $\{x \in X : |h(x)| > t\}$ is in $\mathcal{S}$ because $|h|$ is $\mathcal{S}$-measurable. Because 0 is a lower bound of T_h, the number $\|h\|_\infty$ is the greatest lower bound of T_h whenever T_h is nonempty. If $h \in \mathcal{L}^\infty(\mu)$, then $\inf T_h < \infty$, which implies that T_h is nonempty.

Suppose $f, g \in \mathcal{L}^\infty(\mu)$ and $\alpha \in \mathbf{F}$. Then $f+g$ and αf are $\mathcal{S}$-measurable by algebraic operations with measurable functions (2.46(a)), applied to the real and imaginary parts of f and g.

Suppose $s \in T_f$ and $t \in T_g$. If $x \in X$ is such that $|f(x)| \le s$ and $|g(x)| \le t$, then $|f(x)+g(x)| \le |f(x)| + |g(x)| \le s + t$. Thus

$$\{x \in X : |f(x) + g(x)| > s + t\} \subseteq \{x \in X : |f(x)| > s\} \cup \{x \in X : |g(x)| > t\}.$$

Both sets on the right have measure 0. Because measure preserves order (2.57(a)) and is countably subadditive (2.58), the set on the left has measure 0. Hence $s+t \in T_{f+g}$, and thus $\|f+g\|_\infty \le s+t$ because $\|f + g\|_\infty$ is a lower bound of T_{f+g}. Taking the infimum over $s \in T_f$ gives $\|f + g\|_\infty \le \|f\|_\infty + t$ for every $t \in T_g$, and taking the infimum over $t \in T_g$ then gives $\|f+g\|_\infty \le \|f\|_\infty + \|g\|_\infty$.

To prove that $\|\alpha f\|_\infty = |\alpha|\,\|f\|_\infty$, first suppose $\alpha = 0$. Then $\alpha f = 0$. Because $\{x \in X : |0| > t\} = \emptyset$ for every $t > 0$, we have $T_0 = (0, \infty)$, whose greatest lower bound is 0. Thus $\|\alpha f\|_\infty = \|0\|_\infty = 0 = |\alpha|\,\|f\|_\infty$.

Now suppose $\alpha \ne 0$. If $t > 0$ and $x \in X$, then $|\alpha f(x)| = |\alpha|\,|f(x)|$, which exceeds $|\alpha|t$ if and only if $|f(x)| > t$ because $|\alpha| > 0$. Thus $\{x \in X : |\alpha f(x)| > |\alpha|t\} = \{x \in X : |f(x)| > t\}$ for every $t > 0$, which implies that $T_{\alpha f} = \{|\alpha|t : t \in T_f\}$; this set is nonempty because T_f is. Hence $|\alpha|\,\|f\|_\infty$ is a lower bound of $T_{\alpha f}$. If d is a lower bound of $T_{\alpha f}$, then $d/|\alpha|$ is a lower bound of T_f, which implies that $d/|\alpha| \le \|f\|_\infty$ and thus $d \le |\alpha|\,\|f\|_\infty$. Hence $|\alpha|\,\|f\|_\infty$ is the greatest lower bound of $T_{\alpha f}$; in other words, $\|\alpha f\|_\infty = |\alpha|\,\|f\|_\infty$.

The inequality $\|f + g\|_\infty \le \|f\|_\infty + \|g\|_\infty$ implies that $\|f + g\|_\infty < \infty$, and the equation $\|\alpha f\|_\infty = |\alpha|\,\|f\|_\infty$ implies that $\|\alpha f\|_\infty < \infty$; thus $\mathcal{L}^\infty(\mu)$ is closed under addition and scalar multiplication. Because $\mathcal{L}^\infty(\mu)$ also contains the function 0, whose essential supremum is 0,

the conditions for a subspace (6.31) show that $\mathcal{L}^\infty(\mu)$ is a subspace of $\mathbf{F}^X$ and thus is a vector space. □

Exercise 4.6 (MIRA 7A.2). Suppose $a \geq 0$, $b \geq 0$, and $1 < p < \infty$. Prove that

$$ab = \frac{a^p}{p} + \frac{b^{p'}}{p'}$$

if and only if $a^p = b^{p'}$ [compare to Young's inequality (7.8)].

Solution 4.6. First suppose $a^p = b^{p'}$, and let c denote this common value. Then $a = c^{1/p}$ and $b = c^{1/p'}$ because $a \geq 0$ and $b \geq 0$. The definition of the dual exponent (7.6) gives $\frac{1}{p} + \frac{1}{p'} = 1$, and thus

$$ab = c^{1/p}c^{1/p'} = c^{1/p+1/p'} = c = \frac{c}{p} + \frac{c}{p'} = \frac{a^p}{p} + \frac{b^{p'}}{p'}.$$

To prove the implication in the other direction, suppose $ab = a^p/p + b^{p'}/p'$. Define $g\colon (0,\infty) \to \mathbf{R}$ by $g(t) = (ta)^p/p + b^{p'}/p' - (ta)b$. Young's inequality (7.8), applied to ta and b, shows that $g(t) \geq 0$ for all $t > 0$, and our assumption gives $g(1) = 0$. Thus g attains its minimum at 1. The function g is differentiable, with $g'(t) = a^p t^{p-1} - ab$ for all $t > 0$, and the derivative of a differentiable function on an open interval equals 0 at each point where the function attains its minimum. Hence $a^p - ab = g'(1) = 0$.

Substituting $a^p = ab$ into our assumption gives $ab = ab/p + b^{p'}/p'$, and thus $b^{p'}/p' = ab(1 - 1/p) = ab/p'$ by 7.6. Hence $b^{p'} = ab = a^p$, as desired. □

Exercise 4.7 (MIRA 7A.5). Suppose that $(X, \mathcal{S}, \mu)$ is a measure space, $1 < p < \infty$, $f \in \mathcal{L}^p(\mu)$, and $h \in \mathcal{L}^{p'}(\mu)$. Prove that Hölder's inequality (7.9) is an equality if and only if there exist nonnegative numbers a and b, not both 0, such that

$$a|f(x)|^p = b|h(x)|^{p'}$$

for almost every $x \in X$.

Solution 4.7. Because $1 < p < \infty$ and $\frac{1}{p} + \frac{1}{p'} = 1$, we have $1 < p' < \infty$. Let $s = \|f\|_p$ and $t = \|h\|_{p'}$, which are finite because $f \in \mathcal{L}^p(\mu)$ and $h \in \mathcal{L}^{p'}(\mu)$. The functions $|f|^p$ and $|h|^{p'}$ are $\mathcal{S}$-measurable because $|f|^p$ is measurable if f is measurable (6.20), as is $|fh| = |f|\,|h|$, a product of measurable functions. By the definition of the p-norm (7.1), $\int |f|^p \, d\mu = s^p$ and $\int |h|^{p'} \, d\mu = t^{p'}$, and Hölder's inequality (7.9) gives $\int |fh| \, d\mu = \|fh\|_1 \leq st$. Thus $|f|^p$, $|h|^{p'}$ and $|fh|$ are in $\mathcal{L}^1(\mu)$.

First suppose $s = 0$. Then $0 \leq \|fh\|_1 \leq st = 0$, and thus 7.9 is an equality. Also, $\big\||f|^p\big\|_1 = s^p = 0$, and the second bullet of the properties of the $\mathcal{L}^1$-norm (3.43) implies that $|f(x)|^p = 0$ for almost every $x \in X$. Thus $a|f(x)|^p = b|h(x)|^{p'}$ for almost every $x \in X$, where $a = 1$ and $b = 0$. Hence both conditions of the exercise hold. In the same way, with $a = 0$ and $b = 1$, both hold if $t = 0$. Thus assume that $s > 0$ and $t > 0$.

Define an $\mathcal{S}$-measurable function $g\colon X \to \mathbf{R}$ by

$$g = \frac{|f|^p}{ps^p} + \frac{|h|^{p'}}{p't^{p'}} - \frac{|fh|}{st}.$$

For each $x \in X$, Young's inequality (7.8), applied to the numbers $|f(x)|/s$ and $|h(x)|/t$, shows that $g(x) \geq 0$, and Exercise 7A.2, applied to the same numbers, shows that $g(x) = 0$ if and only

if $|f(x)|^p/s^p = |h(x)|^{p'}/t^{p'}$. Because $g \geq 0$ and the three terms of g are in $\mathcal{L}^1(\mu)$, additivity of integration (3.21) and homogeneity of integration (3.20) give

$$\|g\|_1 = \int g \, d\mu = \frac{s^p}{ps^p} + \frac{t^{p'}}{p't^{p'}} - \frac{\|fh\|_1}{st} = 1 - \frac{\|fh\|_1}{st}.$$

Thus $g \in \mathcal{L}^1(\mu)$, and the equation above with the second bullet of 3.43 shows that $\|fh\|_1 = st$ if and only if $g(x) = 0$ for almost every $x \in X$. Hence 7.9 is an equality if and only if $|f(x)|^p/s^p = |h(x)|^{p'}/t^{p'}$ for almost every $x \in X$.

If 7.9 is an equality, then the equation $a|f(x)|^p = b|h(x)|^{p'}$ holds for almost every $x \in X$ with the positive numbers $a = 1/s^p$ and $b = 1/t^{p'}$. To prove the other direction, suppose a and b are nonnegative numbers, not both 0, such that $a|f(x)|^p = b|h(x)|^{p'}$ for almost every $x \in X$. Let $u = a|f|^p - b|h|^{p'}$, which is in $\mathcal{L}^1(\mu)$ because $|f|^p$ and $|h|^{p'}$ are; then $u(x) = 0$ for almost every $x \in X$. Thus $\|u\|_1 = 0$ by the second bullet of 3.43, and hence $\int u \, d\mu = 0$, because the absolute value of an integral is at most the integral of the absolute value (3.23). By 3.21 and 3.20, $\int u \, d\mu = as^p - bt^{p'}$. Thus $as^p = bt^{p'}$, and this common value, which we call c, is positive because $s > 0$, $t > 0$, and a and b are not both 0. Because $a/c = 1/s^p$ and $b/c = 1/t^{p'}$, dividing both sides of the equation $a|f(x)|^p = b|h(x)|^{p'}$ by c gives $|f(x)|^p/s^p = |h(x)|^{p'}/t^{p'}$ for almost every $x \in X$. Thus 7.9 is an equality by the equivalence proved in the previous paragraph, completing the proof. □

Exercise 4.8 (MIRA 7B.4). Suppose $(X, \mathcal{S}, \mu)$ is a σ-finite measure space and $1 \leq p \leq \infty$. Prove that if $f \colon X \to \mathbf{F}$ is an $\mathcal{S}$-measurable function such that $fh \in \mathcal{L}^1(\mu)$ for every $h \in \mathcal{L}^{p'}(\mu)$, then $f \in \mathcal{L}^p(\mu)$.

Solution 4.8. First suppose $p = 1$. Then $p' = \infty$, and the constant function 1 is in $\mathcal{L}^\infty(\mu)$ because $\|1\|_\infty \leq 1$ by the definition of the essential supremum (7.1). Thus the hypothesis with $h = 1$ shows that $f \in \mathcal{L}^1(\mu)$.

Now suppose $p = \infty$, which gives $p' = 1$, and suppose $f \notin \mathcal{L}^\infty(\mu)$. Then $\mu(\{x \in X : |f(x)| > t\}) > 0$ for every $t > 0$, because otherwise 7.1 would give $\|f\|_\infty \leq t < \infty$ for some $t > 0$. By the definition of a σ-finite measure (5.18), there exist $X_1, X_2, \ldots \in \mathcal{S}$ such that $X = \bigcup_{k=1}^\infty X_k$ and $\mu(X_k) < \infty$ for every $k \in \mathbf{Z}^+$. For $n \in \mathbf{Z}^+$, let $A_n = \{x \in X : |f(x)| > 2^n\}$, which is in $\mathcal{S}$ because f is $\mathcal{S}$-measurable. Because $\mu(A_n) > 0$ and $A_n = \bigcup_{k=1}^\infty (A_n \cap X_k)$, countable subadditivity (2.58) shows that $\mu(A_n \cap X_k) > 0$ for some $k \in \mathbf{Z}^+$; let $B_n = A_n \cap X_k$ for one such k. Then $B_n \in \mathcal{S}$, and $0 < \mu(B_n) \leq \mu(X_k) < \infty$ because measure preserves order (2.57(a)).

For $m \in \mathbf{Z}^+$, let $h_m = \sum_{n=1}^m \chi_{B_n}/\big(2^n \mu(B_n)\big)$. Each $x \in X$ lies in B_n for only finitely many n, because $x \in B_n$ implies that $n < 2^n < |f(x)|$. Thus $h(x) = \lim_{m\to\infty} h_m(x)$ defines a function $h \colon X \to [0, \infty)$, which is $\mathcal{S}$-measurable because a limit of $\mathcal{S}$-measurable functions is $\mathcal{S}$-measurable (2.48). The integral of a linear combination of characteristic functions (3.15) gives $\int h_m \, d\mu = \sum_{n=1}^m 2^{-n} < 1$. Because $0 \leq h_1 \leq h_2 \leq \cdots$, the Monotone Convergence Theorem (3.11) gives $\int h \, d\mu = \lim_{m\to\infty} \int h_m \, d\mu \leq 1$. Because $|f| > 2^n$ on B_n, for each $m \in \mathbf{Z}^+$ we have $|fh| \geq |f| h_m \geq \sum_{n=1}^m \chi_{B_n}/\mu(B_n)$ on X, and the integral of the right side is m by 3.15. Because integration is order preserving (3.8), we have $\int |fh| \, d\mu \geq m$ for every $m \in \mathbf{Z}^+$, and hence $\int |fh| \, d\mu = \infty$. Thus $h \in \mathcal{L}^1(\mu)$ and $fh \notin \mathcal{L}^1(\mu)$, which contradicts the hypothesis. Hence $f \in \mathcal{L}^\infty(\mu)$.

Finally, suppose $1 < p < \infty$, and suppose $f \notin \mathcal{L}^p(\mu)$, which means that $\int |f|^p \, d\mu = \infty$. With $X_1, X_2, \ldots$ as in the previous case, let $E_0 = \emptyset$ and $E_k = \{x \in X_1 \cup \cdots \cup X_k : |f(x)| \leq k\}$

for $k \in \mathbf{Z}^+$, and let $I_k = \int \chi_{E_k} |f|^p \, d\mu$ for each integer $k \geq 0$. Then $E_k \in \mathcal{S}$ and $\mu(E_k) \leq \mu(X_1) + \cdots + \mu(X_k) < \infty$ by 2.57(a) and 2.58. Because $\chi_{E_k}|f|^p \leq k^p \chi_{E_k}$, 3.8 and 3.15 give $I_k \leq k^p \mu(E_k) < \infty$. Furthermore, $E_1 \subseteq E_2 \subseteq \cdots$ and $\bigcup_{k=1}^{\infty} E_k = X$, because each $x \in X$ lies in X_j for some $j \in \mathbf{Z}^+$ and thus in E_k for every integer $k \geq \max\{j, |f(x)|\}$. Hence the functions $\chi_{E_k}|f|^p$ increase pointwise to $|f|^p$, and 3.11 gives $\lim_{k\to\infty} I_k = \int |f|^p \, d\mu = \infty$.

Because each I_k is finite and $\lim_{k\to\infty} I_k = \infty$, induction on n gives integers $0 = k_0 < k_1 < k_2 < \cdots$ such that $I_{k_n} \geq I_{k_{n-1}} + 2^{n(p-1)}$ for every $n \in \mathbf{Z}^+$. For $n \in \mathbf{Z}^+$, let $D_n = E_{k_n} \setminus E_{k_{n-1}}$ and $a_n = \int \chi_{D_n} |f|^p \, d\mu$. The sets $D_1, D_2, \ldots$ are in $\mathcal{S}$ and are disjoint, because $D_n \subseteq E_{k_n} \subseteq E_{k_{m-1}}$ and $D_m \cap E_{k_{m-1}} = \emptyset$ whenever $n < m$. Because $\chi_{E_{k_n}} = \chi_{E_{k_{n-1}}} + \chi_{D_n}$, additivity of integration (3.16) gives $I_{k_n} = I_{k_{n-1}} + a_n$, and thus $2^{n(p-1)} \leq a_n \leq I_{k_n} < \infty$.

Let $h = \sum_{n=1}^{\infty} a_n^{-1} \chi_{D_n} |f|^{p-1}$. Because the sets D_n are disjoint, at most one term of this sum is nonzero at each point of X. Thus h is a function from X to $[0, \infty)$, and h is $\mathcal{S}$-measurable, being the pointwise limit of the partial sums. Because $(p-1)p' = p$ by the definition of the dual exponent (7.6), we have

$$h^{p'} = \sum_{n=1}^{\infty} a_n^{-p'} \chi_{D_n} |f|^p \qquad \text{and} \qquad |fh| = \sum_{n=1}^{\infty} a_n^{-1} \chi_{D_n} |f|^p$$

on X. By homogeneity of integration (3.20), the nth term of the first sum has integral $a_n^{1-p'}$ and the nth term of the second sum has integral 1. Now $1 - p' = -1/(p-1)$ and $a_n \geq 2^{n(p-1)}$, and thus $a_n^{1-p'} \leq 2^{-n}$. For each $m \in \mathbf{Z}^+$, 3.16 shows that the first m terms of the first sum have integral at most $\sum_{n=1}^{m} 2^{-n} < 1$ and the first m terms of the second sum have integral m. Because these partial sums increase pointwise to $h^{p'}$ and to $|fh|$, 3.11 gives $\int h^{p'} \, d\mu \leq 1$ and $\int |fh| \, d\mu = \lim_{m\to\infty} m = \infty$. Thus $h \in \mathcal{L}^{p'}(\mu)$ and $fh \notin \mathcal{L}^1(\mu)$. This contradiction completes the proof. □

Exercise 4.9 (MIRA 7B.5).

(a) Prove that if μ is a measure, $1 < p < \infty$, and $f, g \in L^p(\mu)$ are such that

$$\|f\|_p = \|g\|_p = \left\| \frac{f+g}{2} \right\|_p,$$

then $f = g$.

(b) Give an example to show that (a) can fail if $p = 1$.

(c) Give an example to show that (a) can fail if $p = \infty$.

Solution 4.9. (a) First suppose $a, b \in \mathbf{F}$, and let $s = |a|$, $t = |b|$, $m = (s+t)/2$ and $r = \operatorname{Re}(a\overline{b})$. The properties of complex conjugates (6.24) give

$$|a \pm b|^2 = (a \pm b)(\overline{a} \pm \overline{b}) = s^2 + t^2 \pm 2r \qquad \text{and} \qquad |a\overline{b}|^2 = (a\overline{a})(b\overline{b}) = s^2 t^2.$$

Because the real part of a complex number is at most its absolute value, $r \leq |a\overline{b}| = st$. Thus $|a+b|^2 + |a-b|^2 = 2s^2 + 2t^2$ and $|a+b|^2 \leq (s+t)^2 = 4m^2$, which implies that $|a+b| \leq 2m$. Because $(p-1)p' = p$, Young's inequality (7.8), applied to s and m^{p-1} and then to t and m^{p-1}, gives

$$s\, m^{p-1} \leq \frac{s^p}{p} + \frac{m^p}{p'} \qquad \text{and} \qquad t\, m^{p-1} \leq \frac{t^p}{p} + \frac{m^p}{p'}.$$

Adding these two inequalities and using $s+t = 2m$ and $1-1/p' = 1/p$ gives $2m^p/p \le (s^p+t^p)/p$. Because $y \mapsto y^p$ is strictly increasing on $[0,\infty)$, this inequality and $|a+b| \le 2m$ show that

$$\left|\frac{a+b}{2}\right|^p \le m^p \le \frac{|a|^p+|b|^p}{2}.$$

Now suppose the left side above equals the right side. Then both inequalities in the display above are equalities; thus $|a+b| = 2m$ and $2m^p = s^p + t^p$. The last equation implies that the sum of the two inequalities given by Young's inequality is an equality, and hence both of them are equalities. Thus Exercise 7A.2 gives $s^p = (m^{p-1})^{p'} = m^p$ and $t^p = m^p$, which implies that $s = t = m$. Hence $|a-b|^2 = 2s^2 + 2t^2 - |a+b|^2 = 4m^2 - 4m^2 = 0$, and thus $a = b$.

Now suppose $(X, \mathcal{S}, \mu)$ is a measure space and $f, g \in \mathcal{L}^p(\mu)$ are such that $\|\tilde{f}\|_p = \|\tilde{g}\|_p = \|(\tilde{f}+\tilde{g})/2\|_p$. Let $h = (f+g)/2$, which is in $\mathcal{L}^p(\mu)$ because $\mathcal{L}^p(\mu)$ is a vector space (7.5). Then the definition of addition and scalar multiplication on $L^p(\mu)$ (7.16) and the definition of the norm on $L^p(\mu)$ (7.17) give $\|f\|_p = \|g\|_p = \|h\|_p$; let c denote this common value, which is finite because $f \in \mathcal{L}^p(\mu)$. Let $u = |f|^p + |g|^p - 2|h|^p$. Because $|f|^p$ is measurable if f is measurable (6.20), u is an $\mathcal{S}$-measurable function from X to $\mathbf{R}$. By the first paragraph with $a = f(x)$ and $b = g(x)$, we have $u(x) \ge 0$ for all $x \in X$, with $u(x) = 0$ only if $f(x) = g(x)$. Because $u + |h|^p + |h|^p = |f|^p + |g|^p$ and these functions are nonnegative, additivity of integration (3.16) and the definition of the p-norm (7.1) give $\int u\,d\mu + 2c^p = 2c^p$, and thus $\int u\,d\mu = 0$ because $c < \infty$. Hence $u \in \mathcal{L}^1(\mu)$ with $\|u\|_1 = 0$, and the second bullet of the properties of the $\mathcal{L}^1$-norm (3.43) shows that $u(x) = 0$ for almost every $x \in X$. Thus $f(x) = g(x)$ for almost every $x \in X$; in other words, $g - f \in \mathcal{Z}(\mu)$. Because $\mathcal{Z}(\mu)$ is a vector space (the third bullet of Example 6.32), the definition of $\tilde{f}$ (7.15) gives $\tilde{f} = \tilde{g}$, as desired.

(b) Let μ be counting measure on $\mathbf{Z}^+$. The only set of μ-measure 0 is the empty set; thus $\mathcal{Z}(\mu) = \{0\}$, and sequences that differ in some term represent different elements of $L^1(\mu)$ and of $L^\infty(\mu)$. Let $f = (1, 0, 0, \dots)$ and $g = (0, 1, 0, 0, \dots)$. Then $(f+g)/2 = (\frac{1}{2}, \frac{1}{2}, 0, 0, \dots)$, and the formula for the p-norm with respect to counting measure (Example 7.2) gives $\|f\|_1 = \|g\|_1 = 1$ and $\|(f+g)/2\|_1 = \frac{1}{2} + \frac{1}{2} = 1$. Thus f and g satisfy the hypothesis of (a) with $p = 1$, but $f \ne g$.

(c) Again let μ be counting measure on $\mathbf{Z}^+$, and let $f = (1, 0, 0, \dots)$ and $g = (1, 1, 0, 0, \dots)$. Then $(f+g)/2 = (1, \frac{1}{2}, 0, 0, \dots)$. Each of these three sequences has 1 as the supremum of the absolute values of its terms, and thus 7.2 gives $\|f\|_\infty = \|g\|_\infty = \|(f+g)/2\|_\infty = 1$, but $f \ne g$. $\square$